Karan Grover

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Germany | LinkedIn | Portfolio | Github

SUMMARY

XR Developer | Unity(C#) | Prototyping & Interactive Experiences

XR developer experienced in Unity, Unreal Engine, VR/AR interaction systems, multiplayer prototypes, and real-time visualization workflows for automotive and industrial applications. Currently working on Unreal Engine-based CAD-to-virtual validation pipelines, Blueprint-to-C++ conversion, and automated vehicle visualization workflows.

SKILLS

Unity (C#) & XR :	Unity, C#, XR Interaction Toolkit, Meta Quest, WebGL, spatial UI, controller input, Fish-Networking, Netcode, Photon PUN 2
VR/AR Interaction Development :	Experienced in building intuitive interaction systems for VR/AR applications, including gesture mechanics, and performance-optimized workflows.
Unreal Engine :	C++, Blueprints, CAD-to-Unreal pipelines, real-time rendering, performance profiling
Digital Twin & XR Development :	Proficient in developing VR-based simulations for industrial use cases using Digital Twin concepts and state-of-the-art XR platforms (AR, VR, MR).

EXPERIENCE

BMW Munich, Germany **VR Developer** **05/2026 – Present**

Project - Automated Virtual Vehicle Validation Pipeline in Unreal Engine

- Developing an Unreal Engine-based CAD-to-Virtual Vehicle Validation Pipeline, converting Blueprint workflows into C++, automating CAD model preparation, vehicle rigging, and animation setup, and extending plugin functionality for real-time vehicle visualization.

Lufthansa Technik Hamburg, Germany **VR Developer** **11/2025 – 04/2026**

Project - Multi-User Flight Simulation (VR) & Cross-Sectional Visualization (AR)

- Built multi-user VR flight simulation with Fish-Networking, implementing real-time interaction systems and optimizing performance for smooth multi-user sessions.
- Working on a stereoscopic XR project in Unity (C#), implementing advanced rendering techniques, spatial visualization systems, and optimizing performance for immersive XR experiences on Meta Quest 3.

Fraunhofer IEM Paderborn, Germany **VR Research Assistant** **04/2025 – 10/2025**

Project - VR Simulation for AI Driven Production Line Optimization

- Prototyped a VR simulation in Unity (C#) using digital twin + soft-sensor concepts, taking features from feasibility to validated demo build for production-line safety scenarios.

- Explored AI-driven optimization approaches for predictive analytics and decision support; documented findings and iterated simulation behavior based on evaluation results.

Bosch Sensortec GmbH Reutlingen, Germany **VR Developer** **11/2024 – 04/2025**

Project - Immersive Industrial Simulation in VR

- Developed immersive XR Applications in Unreal Engine using C++ and Blueprints, creating user-centric UI & XR setup support (menus, splash screens, on-boarding interaction).
- Created an automated pipeline that converts CAD files into .uasset format for Instant VR Visualization, Optimized 3D Assets (FBX, STEM, Blender) and Assisted with XR Headset Setup & Troubleshooting in Industrial Simulations.

Fraunhofer IPMS Dresden, Germany **Research Assistant** **10/2023 – 10/2024**

Project - Blood Sugar Concentration Emulation (3d Modelling)

- Designed and modeled 3D-printed setup using Blender to emulate blood sugar sensors, preprocessed signal data and developed predictive ML models with 75% accuracy.
- Documented and presented insights to stakeholders, supporting research validation and decision-making.

PROJECTS

Automotive VR Driving Simulation - Vehicle Experience in Unity

- **Prototyped** a showroom-style VR driving demo prototype in Unity (C#) from scratch.
- **Implemented** car selection and in-car driving interactions inside a custom built London city map.

In-Car VR Keyboard & Input Evaluation in Unity

- **Built** an in-car VR text-entry keyboard in Unity (C#), implementing ray-based selection and direct poke typing interactions.
- **Instrumented** the prototype to log typing speed and error rate, and added a basic comfort/sickness self-check to compare input modes (ray vs poke).

Earth Interaction (VR) – Cinematic Space Navigation & Spatial UI

- **Built** an interactive VR space-to-Earth experience with globe selection and spatial UI pop-ups for information display.
- **Implemented** scaling and navigation interactions with a focus on usability and smooth runtime performance.

Multiplayer Unity Prototype (Netcode + NavMesh + Ragdoll Physics)

- **Built** a real-time multiplayer prototype in Unity (C#) using Netcode, implementing networked player movement, core gameplay synchronization, and AI navigation driven by NavMesh.
- **Implemented** ragdoll physics and synchronized gameplay interactions for stable multi-user sessions.

EDUCATION

MSc, Artificial Intelligence **2027**

Brandenburgische Technishche Universitat, Cottbus
subjects: Virtual reality & agents, Learning in Real and Virtual Humans, data exploration,mathematics, neuroscience, sensory motor etc

Bachelor in Technology, Computer Science *Dehradun* **2022**

Institute of Technology, Dehradun subjects: Data structures and algorithms, mathematics, database management systems, operating systems etc.